

Striatal Crosstalk Between Dopamine and Serotonin Systems

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eLife Assessment

This **useful** study employs optogenetics, genetically-encoded dopamine and serotonin sensors, and patch-clamp electrophysiology to investigate modulations of neurotransmitter release between striatal dopamine and serotonin neurons - a topic of interest to neuroscientists studying the basal ganglia. The results suggest that the dopamine and serotonin systems operate largely in parallel, with the activation of serotonin neurons resulting in a small, transient dopamine release. The authors suggest that this interaction occurs via glutamate release in the ventral tegmental area, findings that are closely related to previous work. Some conclusions are **incomplete** requiring larger samples-sizes and controls.

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Abstract

Dopamine (DA) and serotonin (5-HT) are neuromodulators in reward processing, decision-making, and motivated behavior. While often viewed as opposing or complementary systems, how DA and 5-HT release integrate in the striatum remains elusive. Using optogenetics, fiber photometry, and slice electrophysiology, we found that ventral tegmental area (VTA) DA neuron stimulation increased DA release without affecting 5-HT release. Dorsal raphe nucleus (DRN) 5-HT neuron activation, on the other hand, induced serotonin release and a transient increase in DA in the NAc, likely via glutamate co-release onto VTA DA neurons during the initial stimulation phase. These findings indicate that DA and 5-HT operate largely independently in the striatum, with selective circuit-dependent interactions. This work refines our understanding of DA-5HT interactions and provides a foundation for future research into their roles in motivated behaviors and neuropsychiatric disorders.

Main Text

DA neurons projecting from the VTA to the striatum [1–3] are involved in various cognitive functions, and DA dysregulation has been associated with several neuropsychiatric and neurodegenerative disorders [4–7]. Tryptophan-hydroxylase-positive (TpH+) neurons

synthesize 5-HT in the dorsal raphe nucleus (DRN) and modulates synaptic transmission in the VTA, NAc and DS. Behavioral and pharmacological studies suggest that 5-HT, acting in concertation with DA, may affect reward-seeking, decision making, associative learning, and motor planning [8, 9].

Two conflicting hypotheses regarding the interaction between the DA and 5-HT systems exist. The opponency hypothesis [10–12] suggests that DA and 5-HT have opposite functions: while DA promotes behavioral activation, 5-HT, by contrast, suppresses behavior. Pharmacological studies indicate DA and 5-HT agonists or antagonists have opposite effects on reward processing [13, 14], feeding behavior [14], and self-stimulation [15]. Recent *in vivo* fiber photometry recordings further support this oppositional dynamic, suggesting that DA and 5-HT exhibit inverse activity patterns in the striatum in response to reward [16] and following choice actions [17]. Moreover, 5-HT can inhibit DA activity in the VTA and substantia nigra, leading to reduced DA release in the striatum [10, 18].

Alternatively, 5-HT could directly enhance DA activity or share properties that facilitate reinforcement processing similar to DA. Local 5-HT infusion into the striatum can increase extracellular DA levels [19]. Additionally, 5-HT can facilitate DA's behavioral effects. For example, selective serotonin reuptake inhibitors (SSRIs) increase 5-HT, and lead to a modulative effect in DA self-stimulation thresholds [20] and enhancement of cocaine's reinforcing effects [21]. In contrast, depletion of 5-HT or its precursor blocks cocaine's rewarding effects [22], and impair d-amphetamine's ability to decrease impulsivity [23]. Furthermore, serotonergic activity in the dorsal raphe interacts with DA, encoding complementary aspects of reward processing [24].

The above-listed discrepancies suggest complex mechanisms underlying DA-5-HT crosstalk that remain largely elusive, underscoring a complex interplay driven by receptor-specific interactions and overlapping projections. For example, 5-HT modulation of DA activity occurs through 5-HT_{1A}, 5-HT_{1B}, and 5-HT_{2A} receptor activation, which enhances DA release in the striatum and prefrontal cortex. In contrast, 5-HT_{2C} receptor activity inhibits DA release in mesolimbic regions [25]. Additionally, lesion of DA neurons decreases 5HT spontaneous firing in DRN, whereas lesion of 5-HT neurons enhanced the firing activity of DA neurons in VTA [26].

Both DA and 5-HT are involved in reinforcement and decision-making in opposing and complementary ways [27]. An imbalance between these systems is implicated in various psychiatric disorders, including depression, schizophrenia, compulsion, and addiction [28–30]. However, these interactions are complex and not simply oppositional or coherent, requiring further investigation into their synaptic crosstalk and plasticity mechanisms.

There is increasing recognition of the functional and anatomical overlap between VTA DA neurons and DRN 5-HT neurons, particularly in the striatum, where both systems have dense projections and colocalizations [16, 31]. However, few studies have addressed the bidirectional interactions between the DA and 5-HT systems in response to specific neuronal stimulation, and the interaction between dorsal raphe (DR) serotonergic neurons and striatal DA transmission remains controversial. To address this gap, we investigated the effects of selective activation of DA and serotonin neurons on both DA and 5-HT release in the striatum. Recent advancements in GPCR sensors and refined optogenetic tools allow for real-time observation of neurotransmitter dynamics, enabling targeted interventions and establishing causal links. By stimulating DA neurons in the VTA and 5-HT neurons in the DRN, we aimed to investigate the reciprocal interactions between these systems and elucidate their crosstalk mechanisms.

Our findings demonstrate that stimulation of DA neurons in the VTA alters DA but not 5-HT release in the striatum, suggesting that the 5-HT system in the striatum operates relatively independently of VTA DA activity. In contrast, stimulation of 5-HT neurons in the DRN impacts mainly 5-HT with a

minor impact on DA levels in the striatum, highlighting the reciprocal regulatory effects of serotonergic activity on DA circuits. These results provide new insights into the striatum's bidirectional interplay between DA and 5-HT systems.

Understanding the mechanisms underlying DA-5-HT crosstalk is critical for advancing our knowledge of neuromodulatory systems and developing therapeutic interventions for disorders characterized by imbalances in these networks. By simultaneously assessing DA and 5-HT release in response to targeted neuronal activation, this study contributes to a more comprehensive understanding of how these systems interact to regulate striatal function.

Results

Activating VTA DA neurons increases DA but not 5HT in the striatum

We tested whether optogenetic stimulation of VTA DA neurons results in DA and/or 5-HT in the striatum. We injected AAV-FLEX-ChrimsonR-tdTomato into the VTA of DAT-Cre mice and AAV-dLight X.XX or AAV-GRAB5HT in the NAc or DS to monitor DA or 5-HT levels in the striatum (**Figure 1B, C**). We found that optogenetic stimulation of VTA DA neurons (**Figure 1A**) evoked a rapid and robust release of DA in NAc and DS, at a level within the same range as DA increase evoked by passive cocaine injection at 15mg/kg (**Figure 1D**).

Conversely, we did not observe any change in 5-HT levels in either NAc or DS in response to VTA DA stimulation. In contrast, cocaine readily increased 5-HT levels in the two striatal regions (**Figure 1G**). These findings contradict the finding that optogenetic stimulation of serotonergic neurons in the DRN led to increased DA concentration in the striatum, as measured by microdialysis [31]. These conflicting results may arise from differences in stimulation protocols, measurement modalities, neuronal subpopulations recruited, or variations in experimental conditions, highlighting the complexity of 5-HT-DA interactions in the striatum. Together, these results suggest that VTA DA neurons modulate DA release but not 5-HT release in the striatum.

Activating DRN SERT neurons increases 5HT and transiently DA in the striatum

Then, we tested whether stimulation of DRN SERT neurons produces a release of DA or 5-HT in the striatum (**Figure 2B**). We tested the effects of DRN SERT optogenetic stimulation. We found optogenetic stimulation of DRN SERT neurons evoked rapid and robust release of 5-HT in DS at a level that is comparable to the DA increase evoked by passive cocaine injection at 15mg/kg (**Figure 2C**). On the contrary, we did not observe any significant change in DA levels in either NAc or DS in response to DRN SERT stimulation. In contrast, cocaine could readily increase 5-HT levels in the two regions (**Figure 2E**).

Some studies have reported that electrical stimulation of the DRN inhibits DA neuron activity, suggesting a suppressive influence on DA release [32–34], whereas other findings report that electrical stimulation of the DRN induces DA release in the NAc [35]. We observed a small and transient increase in DA levels in NAc after DR SERT stimulation (**Figure 2C**, insert), significantly different from the baseline 5 seconds after the stimulation (**Figure 2D**).

Glutamate-5HT co-release of DR afferents in the VTA

To further investigate this transient crosstalk, we examined the synaptic afferents onto midbrain and striatal neurons in the acute slice preparation, as DR projections have been claimed to release glutamate in addition to serotonin (REF). Combining optogenetic projection targeting and whole-

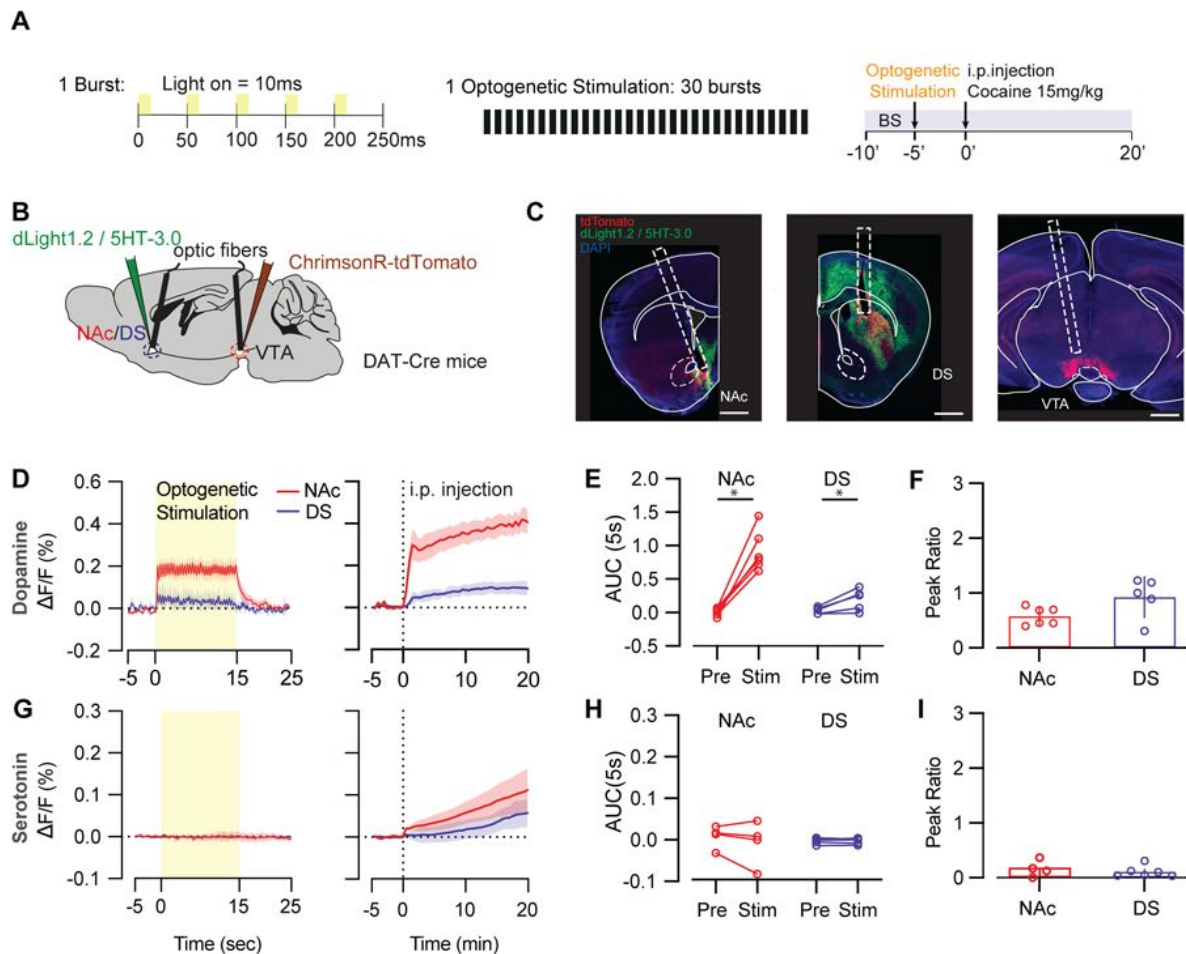


Fig 1.

Stimulation of VTA DA system increases DA but not 5-HT level in the striatum.

(A) Protocol of optogenetic stimulations and cocaine treatments. (B) Schematic of virus injection sites and optic fiber implantation sites for VTA stimulation in DAT-cre mice. (C) (Left) dLight expression and photometry fiber placement in the NAc co-labeled with tdTomato projection from VTA. (Middle) dLight expression and photometry fiber placement in the DS co-labeled with tdTomato projections from VTA. (Right) Optogenetic stimulation fiber placement and Chrimson-tdT expression in the VTA. Scale bars, 1 mm. (D) DA increases after optogenetic stimulation of VTA DA neurons (left) and i.p. cocaine injections (right) in NAc (red trace, $n = 6$) and DS (blue trace, $n = 5$). (E) Area under curve of DA signal in NAc and DS in 5 seconds during baseline and after optogenetic stimulation (Wilcoxon test; NAc: $W = 21$, $z = 0.765$, $*P = 0.016$, $n = 6$ mice; DS: $W = 15$, $z = 0.198$, $*P = 0.031$; $n = 5$ mice respectively). (F) Ratio of peak DA signal after optogenetic stimulation and peak DA signal after cocaine injection. (G) 5-HT increases after optogenetic stimulation of VTA DA neurons (left) and i.p. cocaine injections (right) in NAc (red trace, $n = 4$) and DS (blue trace, $n = 6$). (H) Area under curve of DA signal in DS in 5 seconds during baseline and after optogenetic stimulation (Wilcoxon test; NAc: $W = 2$, $z = -0.01$, $P = 0.375$, $n = 3$ mice; DS: $W = 3$, $z = 0.00$, $P = 0.422$, $n = 6$ mice respectively). (I) Ratio of peak 5-HT signal after optogenetic stimulation and peak 5-HT signal after cocaine injection.

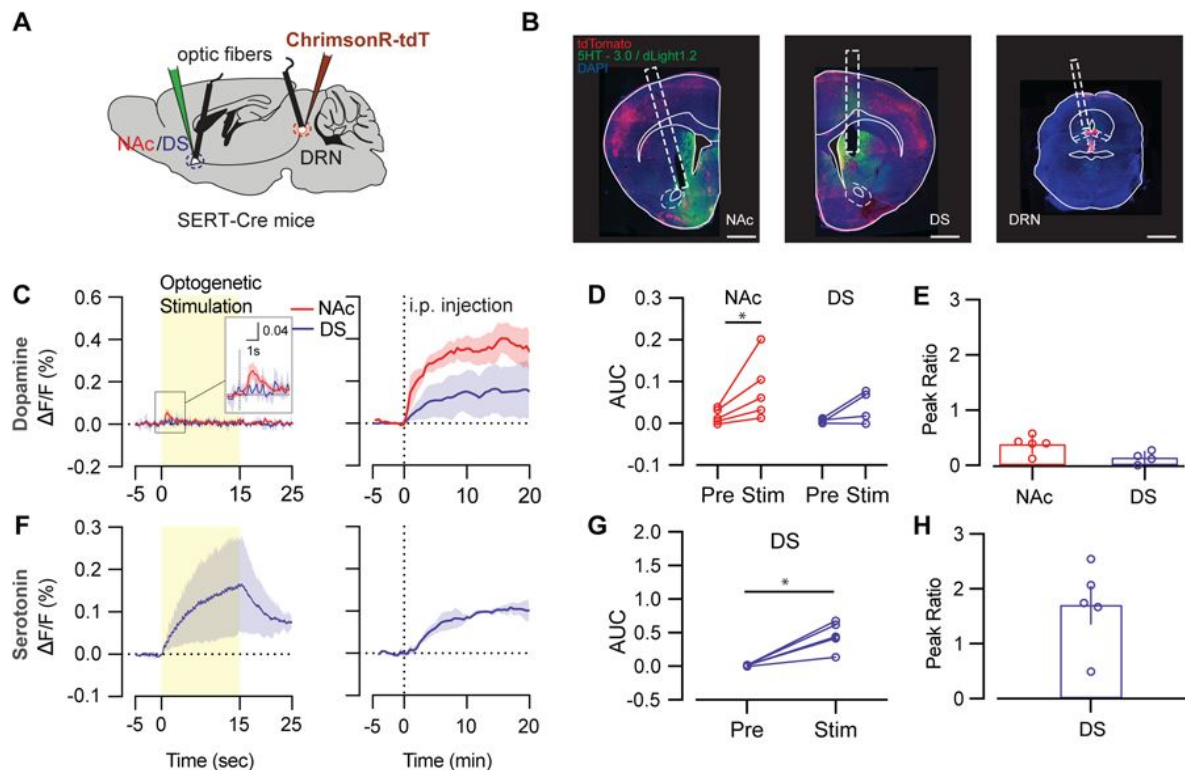


Fig 2.

Stimulation of DRN 5-HT system increases 5-HT but not DA level in the striatum.

(A) Schematic of virus injection sites and optic fiber implantation sites for DR stimulation in SERT-cre mice. (B) (Left) GRAB-5HT expression and photometry fiber placement in the NAc co-labeled with tdTomato projection from DR. (Middle) GRAB-5HT expression and photometry fiber placement in the DS co-labeled with tdTomato projections from DR. (Right) Optogenetic stimulation fiber placement and Chromson-tdT expression in the DR. Scale bars, 1 mm. (C) DA increases after optogenetic stimulation of DR SERT neurons (left) and i.p. cocaine injections (right) in NAc (red trace, $n = 4$) and DS (blue trace, $n = 4$). (D) The area under the curve of DA signal in NAc and DS in 5 seconds during baseline and after optogenetic stimulation. (Wilcoxon test; NAc: $W = 15$, $z = 0.05$, $*P = 0.031$, $n = 5$ mice; DS: $W = 8$, $z = 0.038$, $P = 0.125$, $n = 4$ mice respectively). (E) Ratio of peak DA signal after optogenetic stimulation and peak DA signal after cocaine injection. (F) 5-HT increases after optogenetic stimulation of DR SERT neurons (left) and i.p. cocaine injections (right) in DS (blue trace, $n = 2$). (G) Area under curve of 5-HT signal in DS in 5 seconds during baseline and after optogenetic stimulation. (Wilcoxon test; DS: $W = 15$, $z = 0.423$, $*P = 0.031$, $n = 4$ mice respectively). (H) Ratio of peak 5-HT signal after optogenetic stimulation and peak 5-HT signal after cocaine injection.

cell recordings (**Figure 3A**) allowed parsing the synaptic currents in identified regions. Glutamatergic innervation was absent in DS (**Figure 3B**) and anterior NAc Shell (**Figure 3C**) but observed in OFC (**Figure 3D**) and posterior NAc Shell (**Figure 3E**), as well as the VTA (**Figure 3F**). These results suggest that DRN SERT neurons modulate 5-HT releases in the striatum and indirectly affect DA release. This might be explained by the 5-HT neurons projecting from the DRN also sending glutamatergic projections to VTA, stimulating DA neurons and causing a brief release of DA in the striatum.

Discussion

Our findings demonstrate that optogenetic stimulation of DRN SERT and VTA DA neurons results in minimal crosstalk between striatal DA and 5-HT, indicating that these neurotransmitter systems operate largely independently. This challenges earlier studies that reported DRN stimulation increases striatal DA [31] — a discrepancy likely due to differences in measurement techniques. Microdialysis combined with HPLC may fail to fully distinguish 5-HT from DA because 5-HT shares similar physicochemical properties and overlapping electrochemical signals with 3-methoxytyramine (3-MT, a metabolite of DA)[36].

We employed advanced GPCR-based sensors with high selectivity and sensitivity to overcome these limitations, ensuring a more accurate assessment of neurotransmitter dynamics [37, 38]. Stimulation of DRN SERT neurons elicited a robust, rapid increase in striatal 5-HT levels—comparable to the DA surge induced by cocaine—yet did not significantly alter DA levels in the NAc or DS, except for a transient increase in the NAc. This modest DA change is consistent with previous findings[35] and may be explained by DRN 5-HT neurons projecting to the VTA and co-release glutamate, which can also excite VTA DA neurons and evoke DA release in NAc.

Moreover, our analysis revealed that the 5-HT-glutamate co-release mechanism was present in the OFC and posterior NAc shell but absent in the DS and anterior NAc shell, underscoring the region-specific effects of DRN SERT activation. Our results refine the understanding of DArgic and serotonergic interactions in the striatum, highlighting their functionally distinct roles. These insights provide a foundation for future research into the interplay between these systems in cognitive and behavioral functions, with implications for neuropsychiatric disorders.

Methods

Animals

DAT-IRES-Cre (B6.SJL-Slc6a3^{tm1.1(cre)Bkmn}/J) and SERT-Cre mice (B6.129(Cg)-Slc6a4^{tm1(cre)}Xz/J) were from the Jackson Laboratory. Upon the arrival of the animals a period of 7 days of habituation was made. Both male and female mice aged from 8-12 weeks were used and group housed in a temperature and humidity-controlled environment, under a 12 h light/dark cycle, and provided with food and water *ad libitum*. Weights and genders were distributed homogeneously among the groups if possible. All behavioral procedures were performed during the light cycle. All procedures were approved by the Institutional Animal Care and Use Committee of the University of Geneva and by the animal welfare committee of the Cantonal of Geneva, in accordance with Swiss law (GE131A_33975).

Virus injection and implantation

Mice (age 8-12 weeks) were deeply anesthetized with a mixture of Isoflurane (w/v, induction 5%, maintenance 2%, Attanetm) and O₂ (Compact anesthesia station from Minerve) during surgery and then secured in a stereotaxic frame (Angle One). Before craniotomy, body temperature was

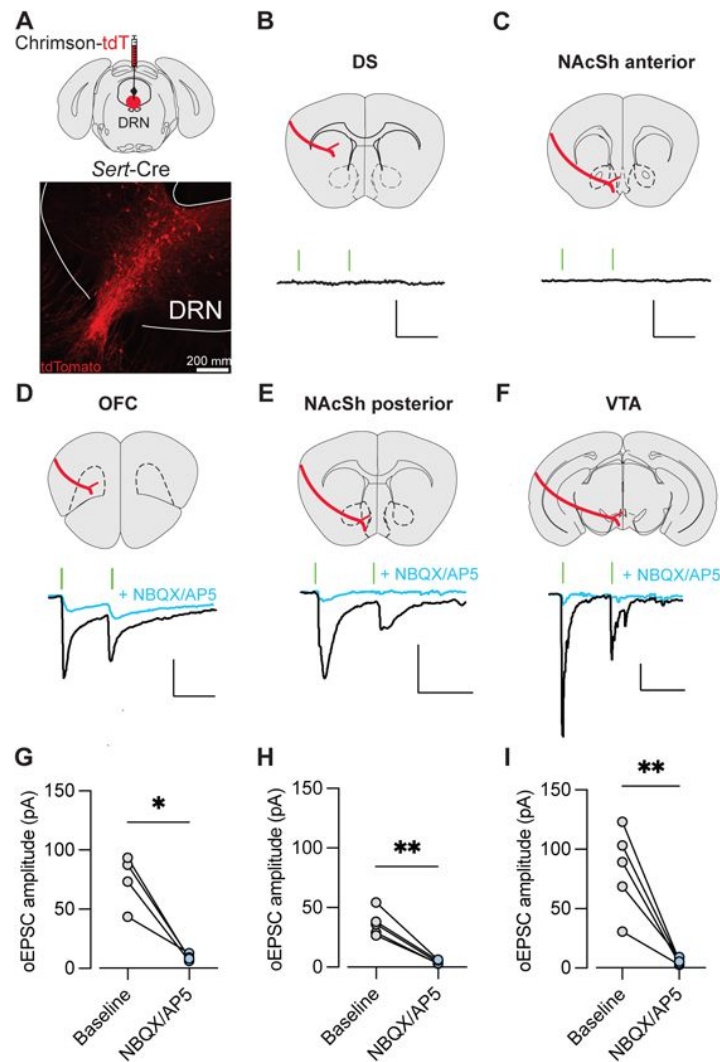


Fig 3.

DRN 5HT neurons make excitatory synapses onto OFC, pNACSh, and VTA neurons.

(A) Top: Diagram of virus injection of AAV-Syn-FLEX-Chrimson-tdTomato in the DRN of SERT-Cre mice. Bottom: Representative image of tdTomato-expressing cell bodies of Sert neurons in the DRN. Scale bar, 200 μ m. (B-F) Top: Localization of recordings in the DS (B), NAcSh anterior (C), OFC (D), NAcSh posterior (E) and VTA (F). Bottom: representative traces of p-pulse optical stimulation of Chrimson+ fibers, before and after bath application of NBQX/AP5 (20 mM) in OFC (C), NAcSh posterior (D) and VTA (E) neurons. (G-I): Quantification of the amplitude of responses before and after NBQX/AP5 bath-application.

maintained at 37 °C with a temperature controller system and Lacryvisc (Alcon, Switzerland) was applied to prevent the eyes from drying. Lidocaine was applied on the surface of the epicranium. An incision was made to expose the bregma and lambda point of the skull. The skull above the target area was thinned with a dental drill and carefully removed. Viruses were injected with glass pipette at a rate of 50 nl/min. The amount of virus per injection site was 350-500 nl. After injection, the pipette was left in the place for 10 min to allow diffusion of the virus. The skin was sutured and disinfected after the injection. 500 µl saline was i.p. injected. Paracetamol (2 mg/ml in the water bottle) was given orally for the next 2-4 days.

(anterior posterior (AP): -3.28; medio-lateral (ML): -0.9; dorso-ventral (DV): -4.3; with a 10° angle)

To express opsins, AAV8-syn-Flex-Chrimson-tdTomato (UNC) was injected in VTA (anterior posterior (AP): -3.28; medio-lateral (ML): -0.9; dorso-ventral (DV): -4.3; with a 10° angle) of DAT-Cre mice or DRN (AP: -4.5; ML: 0; DV: -3) of SERT-Cre mice. To record DA release, an AAV5-CAG-dLight1.2 (400 nl, from Addgene) was injected in NAc (AP: +1.6; ML: ±0.75; DV: -4.3) and DS (AP: +1.3; ML: ±0.1; DV: -3.3). To record 5-HT release, AAV9-hSyn-5-HT3.0 (400 nl, from BrainVTA) was injected in NAc (AP: +1.6; ML: ±0.75; DV: -4.3) and DS (AP: +1.3; ML: ±0.1; DV: -3.3).

During the same surgical procedure, three screws were fixed into the skull to secure the optical implant. An optic fiber (0.4 mm diameter, MFC_400/430_0.48_4mm_ZF2.5(G)FLT, Doric Lenses) was implanted 150 µm above the virus injection site for in vivo fiber photometry recording. Another optic fiber (0.2 mm diameter, FOC-W-1.25-200-0.37-5.0, Inper) was implanted 200 µm above the virus injection site for optogenetic stimulation. Then the optic fibers were fixed to the skull with dental cement. After surgery mice were given 7 days of recovery and were habituated to handling.

For electrophysiological studies examining DRN SERT connectivity, AAV8-syn-Flex-Chrimson-tdTomato (UNC) was injected in DRN (AP: -4.5; ML: 0; DV: -3) of SERT-Cre mice.

Behavioral Paradigm

After 4-5 week of viral expression, mice were habituated to handling and to the connection cable for three day before testing. On the testing day mice were connected to the optogenetics cable and fiber photometry cable and placed in a transparent custom build open-field (30 X 30 X 20 cm) for 3 min of habituation before the start of recording. Then, 5 min of baseline recording were made. Animal received 5 pulses of laser stimulations, each composed of 30 bursts separated by 250 ms (each burst consisted of 5 laser pulses of 4-ms pulse width at 20 Hz). After another 5 minutes of baseline recording, animals received i.p. injection of cocaine (15mg/kg), and the neuronal activity was recorded for another 40 minutes.

Optogenetic stimulation

Optic fiber for optogenetic stimulation of mice were connected via patch cords (MFO-F- W1.25-200-0.37-100, Inper) to a rotary joint (FRJ_1 × 2_FC-2FC; Doric Lenses, Quebec, Canada), suspended above the operant chamber. A second patch cord was connected from the rotary joint to an orange DPSS laser (SDL-593-100 mW; Shanghai Dream Lasers; Shanghai, China) positioned outside of the context. Laser power was typically 15-20 mW measured at the end of each patch cord. A mechanical shutter was used to control laser output (SR474 driver with SR476 shutter head; Stanford Research Systems, aligned using a connectorized mechanical shutter adaptor; Doric Lenses).

Fiber Photometry recordings

Fiber photometry was performed similar to before (33). During recordings, excitation (470 nm, M470F3, Thorlabs) and control LED light (405 nm, M405FP1, Thorlabs) was passed through excitation filters and focused onto a patch cord. Both LED lights were sinusoidally modulated at

211 and 531 Hz (470 nm and 405 nm light, respectively) and light was passed through a mini cube (FMC4_AE(405)_E(450-490)_F(500-550)_S) onto the optic fiber patch cable (MFP_400/430/1100-0.48_4 m_FC-ZF2.5, Doric Lenses) that was connected to the chronically implanted fiber (MFC_400/430-0.48_6mm_ZF2.5(G)_FLT, Doric Lenses). Light intensity at the tip of the patch cable was around 0.25 mW. Emission light travelled back through the same fibers onto the mini cube and a photo-receiver (Newport 2151, Doric Lenses), after which it was digitized, demodulated and stored using a signal processor (RZ5P, Tucker Davis Technologies).

The sample rate was 101.725 Hz, signals were low-pass filtered online at 3 Hz.

The data were analyzed using MATLABR2024 (MathWorks). First the signal during baseline acquisition originating from the 405 nm excitation source was linearly regressed to the signal originating from the 470 nm excitation source, and scale to the 470 nm originating signal. $\Delta F/F$ was then computed as (470 nm signal – fitted 405 nm signal)/fitted 405 nm signal. The average $\Delta F/F$ signal in the baseline periods before experimental intervention was then subtracted to normalize the signal to baseline. Finally, the normalized signal was binned into appropriate time bins in the graphs and analyses.

Electrophysiology recording ex vivo

SERT-Cre mice were injected with AAV-FLEX-ChR2som-tdT in the DRN to detect potential glutamate corelease from 5-HT neurons. Three weeks after injection, mice were deeply anesthetized with 5% isoflurane and subsequently decapitated after confirmation of absent of toe and tail reflexes.

92 mM NMDG, 20 mM HEPES, 25 mM glucose, 30 mM NaHCO₃, 2.5 mM KCl, 1.2 mM NaH₂PO₄ (pH 7.35, 303–306 mOsm) and saturated with 95% O₂/5% CO₂. Coronal slices (300 μm) containing the OFC, DS, NAc anterior, NAc posterior and VTA were prepared at a speed of 0.07 mm/s using a vibratome (Leica, VT1200). After slicing, sections were allowed to recover at 34°C in bubbled NMDG-based for 5-10 min, before being transferred to a holding chamber filled with modified holding aCSF saturated with 95% O₂/5% CO₂ containing 92 mM NaCl, 20 mM HEPES, 25 mM glucose, 30 mM NaHCO₃, 2.5 mM KCl, 1.2 mM NaPO₄ (pH 7.35, 303–306 mOsm) at room temperature for at least 1 h.

For recordings, slices were kept at 30 °C in a recording chamber perfused with 1.5 – 2.0 ml/min ACSF containing 126 mM NaCl, 2.5 mM KCl, 1.4 mM NaH₂PO₄, 1.2 mM MgCl₂, 2.4 mM CaCl₂, 25 mM NaHCO₃ and 11 mM glucose (303– 305 mOsm). Neurons were visualized with a ×40 water-immersion objective on an Olympus BX51WI inverted microscope equipped with infrared-differential interference contrast (IR- DIC) optics and epifluorescence (Olympus Corp). Td-tomato-positive cells were classified based on strong fluorescence, while td-tomato-negative cells were those adjacent to td- tomato-positive cells, but lacking fluorescence. Whole cell recordings were amplified (Multiclamp 700B, Axon Instruments), filtered at 5 kHz and digitized at 20 kHz (National Instruments Board PCI-MIO-16E4, Igor, Wave Metrics). Data were rejected if the access resistance changed more than 20%. Optogenetic EPSCs were evoked with single or paired pulses of 0.1 Hz, with an interstimulus interval of 50 ms.), and recorded in the presence of PTX (100 μM, Tocris).

Histological analysis

Brains were fixed with 4% paraformaldehyde (PFA) for at least 24 h and sliced with vibratome (Leica, VT1200). 50 μm thick slices were washed with 3 times PBS for 5 min and blocked with 10% bovine serum albumin (BSA) dissolved in 0.5% TritonX-100 for 1 hour at room temperature. Slices were incubated with primary antibodies dissolved in blocking buffer overnight at 4 °C, followed by 4 times 15 min wash with PBS at room temperature.

After that, slices were incubated with fluorescent-conjugated secondary antibodies dissolved in blocking buffer for 2 h at room temperature followed by 4 times 15 min wash with PBS. Last, slices were mounted using mounting medium containing DAPI (Fluoroshield, Abcam). Primary and secondary antibodies are listed below: Rabbit anti GFP (1:500, Invitrogen, A11122, RRID: AB_221569); Goat anti Rabbit 488 (1:500, Invitrogen, A11008, RRID: AB_143165).

Statistical analysis

Statistical analysis was performed in GraphPad Prism 9. For all tests, the significance threshold was placed at $\alpha = 0.05$. Gaussian distribution was evaluated using D'Agostino & Pearson normality test. Multiple comparisons were first subject to mixed-factor ANOVA defining both between- and/or within-group factors. Where significant main effects or interaction between factors were found ($P < 0.05$), further comparisons were made by a two-tailed Student's t-test with Bonferroni corrections applied when appropriate (that is, the level of significance equaled 0.05 divided by the number of comparisons). Mann-Whitney or Friedman test were used for non-Gaussian distributions when appropriate. Single comparisons of between- or within-group measures were made by two-tailed non-paired or paired Student's t-test, respectively.

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References

1. Robinson T.E., Berridge K.C. (1993) **The neural basis of drug craving: an incentive-sensitization theory of addiction** *Brain research reviews* **18**:247–291 [Google Scholar](#)
2. Horvitz J.C. (2000) **Mesolimbocortical and nigrostriatal dopamine responses to salient non-reward events** *Neuroscience* **96**:651–656 [Google Scholar](#)
3. Haber S.N., Fudge J.L., McFarland N.R. (2000) **Striatonigrostriatal pathways in primates form an ascending spiral from the shell to the dorsolateral striatum** *J Neurosci* **20**:2369–82 [Google Scholar](#)
4. Di Chiara G., Imperato A. (1988) **Drugs abused by humans preferentially increase synaptic dopamine concentrations in the mesolimbic system of freely moving rats** *Proceedings of the National Academy of Sciences* **85**:5274–5278 [Google Scholar](#)
5. Schultz W. (2007) **Multiple dopamine functions at different time courses** *Annu. Rev. Neurosci* **30**:259–288 [Google Scholar](#)
6. Ikemoto S. (2010) **Brain reward circuitry beyond the mesolimbic dopamine system: A neurobiological theory** *Neuroscience & Biobehavioral Reviews* **35**:129–150 [Google Scholar](#)
7. Pascoli V., et al. (2015) **Sufficiency of mesolimbic dopamine neuron stimulation for the progression to addiction** *Neuron* **88**:1054–1066 [Google Scholar](#)
8. Lucki I. (1998) **The spectrum of behaviors influenced by serotonin** *Biol Psychiatry* **44**:151–62 [Google Scholar](#)
9. Barton D.A., et al. (2008) **Elevated brain serotonin turnover in patients with depression: effect of genotype and therapy** *Archives of General Psychiatry* **65**:38–46 [Google Scholar](#)
10. Daw N.D., Kakade S., Dayan P. (2002) **Opponent interactions between serotonin and dopamine** *Neural networks* **15**:603–616 [Google Scholar](#)
11. Rogers R.D. (2011) **The Roles of Dopamine and Serotonin in Decision Making: Evidence from Pharmacological Experiments in Humans** *Neuropsychopharmacology* **36**:114–132 [Google Scholar](#)
12. Boureau Y.L., Dayan P. (2011) **Opponency Revisited: Competition and Cooperation Between Dopamine and Serotonin** *Neuropsychopharmacology* **36**:74–97 [Google Scholar](#)
13. Fletcher P.J., Ming Z.-H., Higgins G.A. (1993) **Conditioned place preference induced by microinjection of 8-OH-DPAT into the dorsal or median raphe nucleus** *Psychopharmacology* **113**:31–36 [Google Scholar](#)
14. Fletcher P.J., Korth K.M., Chambers J.W. (1999) **Selective destruction of brain serotonin neurons by 5,7-dihydroxytryptamine increases responding for a conditioned reward** *Psychopharmacology* **147**:291–299 [Google Scholar](#)

15. Fletcher P.J., Tampakeras M., Yeomans J.S. (1995) **Median raphe injections of 8-OH- DPAT lower frequency thresholds for lateral hypothalamic self-stimulation** *Pharmacology Biochemistry and Behavior* **52**:65–71 [Google Scholar](#)
16. Cardozo Pinto D.F., et al. (2024) **Opponent control of reinforcement by striatal dopamine and serotonin** *Nature* [Google Scholar](#)
17. Bang D., et al. (2020) **Sub-second Dopamine and Serotonin Signaling in Human Striatum during Perceptual Decision-Making** *Neuron* **108**:999–1010 [Google Scholar](#)
18. Kapur S., Remington G. (1996) **Serotonin-dopamine interaction and its relevance to schizophrenia** *Am J Psychiatry* **153**:466–76 [Google Scholar](#)
19. De Deurwaerdère P., et al. (1996) **Serotonin enhances striatal dopamine outflow in vivo through dopamine uptake sites** *J Neurochem* **66**:210–5 [Google Scholar](#)
20. Harrison A.A., Markou A. (2001) **Serotonergic Manipulations Both Potentiate and Reduce Brain Stimulation Reward in Rats: Involvement of Serotonin-1A Receptors** *The Journal of Pharmacology and Experimental Therapeutics* **297**:316–325 [Google Scholar](#)
21. Sasaki-Adams D.M., Kelley A.E. (2001) **Serotonin-Dopamine Interactions in the Control of Conditioned Reinforcement and Motor Behavior** *Neuropsychopharmacology* **25**:440–452 [Google Scholar](#)
22. Aronson S.C., et al. (1995) **Serotonergic mechanisms of cocaine effects in humans** *Psychopharmacology* **119**:179–185 [Google Scholar](#)
23. Winstanley C.A., et al. (2005) **Interactions between Serotonin and Dopamine in the Control of Impulsive Choice in Rats: Therapeutic Implications for Impulse Control Disorders** *Neuropsychopharmacology* **30**:669–682 [Google Scholar](#)
24. Nakamura K., Matsumoto M., Hikosaka O. (2008) **Reward-Dependent Modulation of Neuronal Activity in the Primate Dorsal Raphe Nucleus** *The Journal of Neuroscience* **28**:5331–5343 [Google Scholar](#)
25. Di Matteo V., et al. (2008) **Serotonin control of central dopaminergic function: focus on in vivo microdialysis studies** In: Di Giovanni G., Di Matteo V., Esposito E, editors. *Progress in Brain Research* Elsevier pp. 7–44 [Google Scholar](#)
26. Guiard B.P., et al. (2008) **Functional interactions between dopamine, serotonin and norepinephrine neurons: an in-vivo electrophysiological study in rats with monoaminergic lesions** *International Journal of Neuropsychopharmacology* **11**:625–639 [Google Scholar](#)
27. Cools R., Nakamura K., Daw N.D. (2011) **Serotonin and Dopamine: Unifying Affective, Activational, and Decision Functions** *Neuropsychopharmacology* **36**:98–113 [Google Scholar](#)
28. Li Y., et al. (2021) **Synaptic mechanism underlying serotonin modulation of transition to cocaine addiction** *Science* **373**:1252–1256 [Google Scholar](#)

29. Pelloux Y., et al. (2012) **Reduced forebrain serotonin transmission is causally involved in the development of compulsive cocaine seeking in rats** *Neuropsychopharmacology* **37**:2505–14 [Google Scholar](#)
30. Koob G.F (1992) **Drugs of abuse: anatomy, pharmacology and function of reward pathways** *Trends Pharmacol Sci* **13**:177–84 [Google Scholar](#)
31. Dagher M., et al. (2022) **Optogenetic Stimulation of Midbrain Dopamine Neurons Produces Striatal Serotonin Release** *ACS Chem Neurosci* **13**:946–958 [Google Scholar](#)
32. Dray A., et al. (1976) **Evidence for the existence of a raphe projection to the substantia nigra in rat** *Brain Research* **113**:45–57 [Google Scholar](#)
33. Chang-Tien T (1989) **Involvement of serotonin in mediation of inhibition of substantia nigra neurons by noxious stimuli** *Brain Research Bulletin* **23**:121–127 [Google Scholar](#)
34. Trent F., Tepper J.M. (1991) **Dorsal raphe stimulation modifies striatal-evoked antidromic invasion of nigral dopaminergic neurons in vivo** *Exp Brain Res* **84**:620–30 [Google Scholar](#)
35. Wang H.L., et al. (2019) **Dorsal Raphe Dual Serotonin-Glutamate Neurons Drive Reward by Establishing Excitatory Synapses on VTA Mesoaccumbens Dopamine Neurons** *Cell Rep* **26**:1128–1142 [Google Scholar](#)
36. Ferry B., et al. (2014) **Analysis of microdialysate monoamines, including noradrenaline, dopamine and serotonin, using capillary ultra-high performance liquid chromatography and electrochemical detection** *Journal of Chromatography B* **951–952**:52–57 [Google Scholar](#)
37. Labouesse M.A., Cola R.B., Patriarchi T. (2020) **GPCR-Based Dopamine Sensors-A Detailed Guide to Inform Sensor Choice for In vivo Imaging** *Int J Mol Sci* **21** [Google Scholar](#)
38. Patriarchi T., et al. (2018) **Ultrafast neuronal imaging of dopamine dynamics with designed genetically encoded sensors** *Science* **360** [Google Scholar](#)

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Reviewer #1 (Public review):

Summary:

In this study, Liu et al use optogenetics and genetically encoded neuromodulator sensors to test the extent to which dopamine neuron stimulation produces striatal serotonin release, and vice versa. The study is timely given growing interest in dopamine/serotonin interactions and in the context of recent work showing bidirectional and dynamic regulation of striatal dopamine by another neuromodulator, acetylcholine. The authors find that striatal dopamine and serotonin afferents function largely independently, with dopamine neuron stimulation producing no striatal serotonin release and serotonin neuron stimulation producing minimal striatal dopamine release. This work will inform future work seeking to dissect the contributions of striatal dopamine, serotonin, and their interactions to various motivated behaviors. While the paper's main conclusions are adequately supported (see Strengths), additional controls and experiments would significantly broaden the paper's impact (see Weaknesses). Finally, this draft of the work is poorly presented with numerous errors, omissions, and inconsistencies evident throughout the text and the figures that should be addressed.

Strengths:

The study employs optogenetic stimulation simultaneously with fiber photometry recording of dopamine or serotonin release measured with genetically encoded sensors. These methods are state-of-the-art, offering tighter temporal control compared to pharmacological methods for manipulating dopamine and serotonin and improved selectivity over techniques like electrochemistry and microdialysis used to record neuromodulator release in previous studies on the subject. As a result, the paper's main conclusions are well supported.

Weaknesses:

(1) The electrophysiology experiments in Figure 3 are only tangentially related to the focus of the study, and their findings are almost entirely irrelevant to the paper's main conclusions. The results of these experiments are also not novel. Glutamate corelease from 5HT neurons has been previously shown, including in the OFC and VTA (Ren et al, 2018, Cell, McDevitt et al, 2014, Cell Rep, Liu et al 2014, Neuron; and others). The authors should explain more clearly what they think these data add to the manuscript and/or consider removing them altogether.

(2) Related to the point above, as far as I can tell, the only value the electrophysiology data add is to suggest that perhaps activation of serotonin neurons may drive minimal striatal dopamine release via glutamate corelease in the VTA. The evidence provided in this version of the manuscript is insufficient to support that claim, but the manuscript would be significantly strengthened if the authors tested this hypothesis more directly. One way to do

that could be to stimulate serotonin axons in the striatum (as opposed to the serotonin cell bodies) and record striatal dopamine release. A complementary anatomical approach would be to use retrograde tracing to test whether the DR 5HT neurons projecting to the striatum are the same or different from the VTA projecting population.

(3) The findings would be strengthened by the addition of a fluorophore-only control group lacking opsin expression in all experiments in Figures 1 and 2.

(4) The experiment of stimulating serotonin neurons and recording serotonin release in the NAc was not performed. It would be useful to be able to compare the magnitudes of evoked serotonin release in these two striatal regions, though it is not central to the main claims of the paper.

(5) The interpretation of the results from Figure 2 is described inconsistently throughout the manuscript. The title implies there is significant crosstalk between the dopamine and serotonin systems. The abstract calls the crosstalk "transient", which is a description of its temporal dynamics, not its magnitude. Then the introduction figures and discussion all suggest the crosstalk is minimal. I suggest the authors describe the main findings - minimal crosstalk between the dopamine and serotonin systems - clearly and consistently in the title, abstract, and main text.

<https://doi.org/10.7554/eLife.107252.1.sa2>

Reviewer #2 (Public review):

Summary:

This brief communication aims to clarify interactions between the dopamine (DA) and serotonin (5HT) systems of mice. The authors use optogenetic stimulation of DA neurons in the VTA or of 5HT neurons in the DRN, while monitoring the fluorescence of DA and 5HT sensors in the nucleus accumbens (NAc) and dorsal striatum (DS) using fiber photometry. The authors report on a small release of DA in the NAc following DRN stimulation, which they attribute to glutamate co-release onto VTA DA neurons using slice electrophysiology. The authors also report on cocaine-induced 5-HT release in the striatum.

Strengths:

This is a topic well worth studying.

Weaknesses:

In its current form, this is an incomplete and underpowered study that does little to clarify the complicated relationship that exists between DA and 5HT in the mammalian brain under physiological conditions or during cocaine use.

<https://doi.org/10.7554/eLife.107252.1.sa1>

Reviewer #3 (Public review):

The authors suggest that the small release of DA may be due to a release of glutamate from DRN 5-HT neurons to the VTA that stimulates weakly and in a transient fashion the VTA DA neurons, which in the end, produce a transient and small release of DA in the NAc.

Their findings give more information on the previously reported complex and partial known crosstalk between 5-HT and DA in the NAc.

I only have some minor concerns about the manuscript:

- (1) In Figure 2F, there is a missing curve for 5-HT in NAc. Besides, the legend shows $n=2$, making it difficult to perform statistical analysis with that data.
- (2) In Figure 3, the use of NBQX/AP5 is shown, but it is not mentioned either in the methodology or in the discussion. What is the meaning of those results?
- (3) Line 98 compares results from two different places of stimulation. The results are related to stimulation in the VTA, but the comparison indicates that the stimulation was made in the DRN.
- (4) If the release of 5-HT in Nac does not occur, it needs to be precise in the abstract that 5-HT is released in the dorsal striatum (DS) but not in the NAc (line 19).
- (5) Be consistent with the way you mention the 5-HT neurons. For example, in lines from 106 to 119, SERT neurons are used. Previously, 5-HT neurons were used.
- (6) There are several points of confusion when referring to the figures, making the text difficult to follow because the text explains something that is not shown in the figure cited.

<https://doi.org/10.7554/eLife.107252.1.sa0>

Author response:

We appreciate the reviewers' insightful feedback and propose to undertake an extensive revision of the manuscript to strengthen our findings and underscore the significance of this work. We remain convinced that our study offers critical insights into the largely independent dopamine and serotonin neural circuits. Nevertheless, we concur that substantial revisions are warranted, as the current organization may not be ideal to showcase the central findings. In particular, we will increase the number of animals to address data variability and enhance the reproducibility of the observed effects. We also recognize the need to perform additional control experiments and to include complementary anatomical tracing studies. Moreover, we will reformat the manuscript and conduct additional analyses to emphasize that evoked dopamine and serotonin release originate from distinct loci with minimal crosstalk. To address all of these points thoroughly, we estimate that a 12-month revision period will be required.

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